1. Ginning is the process of separating:

A. Seeds from fibre by dissolving them

B. Lint from seeds without damage to the fibre

C. Trash from lint using air only

D. Fibres into tufts

Ans: B

2. Approximate production of a roller gin is:

A. 250 kg/hr

B. 500 kg/hr

C. 1000 kg/hr

D. 2000 kg/hr

Ans: A

3. Sea Island, Egyptian and Pima cottons are preferably ginned on:

A. Saw gin

B. Rotor gin

C. Roller or McCarthy gin

D. Any gin equally

Ans: C

4. The "cake cotton" fault is a disadvantage of:

A. Saw ginning

B. Roller ginning

C. Carding

D. Combing

Ans: B

5. In a saw gin, lint is removed from the saw teeth by:

A. Stationary knife

B. Seed grid

C. Doffer

D. High-speed brushes or air blast

Ans: D

Opening, Cleaning, Blowroom

6. In the beater and grid bar action, heavy trash is released by:

A. Suction only

B. Centrifugal force and impact

C. Water spray

D. Opposing spikes

Ans: B

7. Blending differs from mixing because blending uses:

A. Random proportions

B. Only waste fibre

C. Precise, measured, controlled proportions

D. Only one variety

Ans: C

8. A step cleaner is inclined at:

A. 30°

B. 45°

C. 60°

D. 90°

Ans: B

9. Cleaning efficiency of the mono cylinder cleaner is about:

A. 5–10%

B. 50–60%

C. 80–90%

D. 25–30%

Ans: D

10. The Uni-Mixer receives material in how many vertical trunks?

A. 4

B. 6

C. 8

D. 12

Ans: C

11. Which is an advantage of the chute feed system?

A. Eliminates lap formation and lap faults

B. Better weighing of individual laps

C. Needs more manpower

D. Blend change gives lower waste

Ans: A

12. The scutcher beater speed is:

A. 100–300 rpm

B. 300–500 rpm

C. 750–1100 rpm

D. 3000–4000 rpm

Ans: C

Carding

13. In carding action, the teeth of the two surfaces face:

A. The same direction

B. Perpendicular directions

C. No direction

D. Opposite directions

Ans: D

14. The cylinder to doffer setting is about:

A. 0.1 mm

B. 1 mm

C. 5 mm

D. 10 mm

Ans: A

15. The percentage of trailing hooks in carded sliver is about:

A. 15%

B. 50%

C. 85%

D. 100%

Ans: B

16. The revolving flat belt contains about:

A. 20–30 bars

B. 150–200 bars

C. 80–116 bars

D. 300 bars

Ans: C

17. The main cylinder diameter of the card is:

A. 250 mm

B. 600 mm

C. 2000 mm

D. 1280–1300 mm

Ans: D

18. Licker-in speed for cotton is:

A. 100–200 rpm

B. 800–1200 rpm

C. 3000–4000 rpm

D. 6000 rpm

Ans: B

19. In the heel and toe arrangement, the flats-to-cylinder setting is wider at:

A. The leading (front) edge

B. The middle

C. The trailing (back) edge

D. All places equally

Ans: C

20. Before transfer to the doffer, fibres rotate on the main cylinder about:

A. 1–2 times

B. 5–10 times

C. 20–30 times

D. 50 times

Ans: B

Draw Frame

21. If CV of fed sliver is 8% and 4 slivers are doubled, CV delivered is:

A. 1%

B. 2%

C. 4%

D. 8%

Ans: C

22. Typical number of slivers doubled at the draw frame is:

A. 1–2

B. 6–8

C. 20–24

D. 60

Ans: B

23. Dust removal efficiency of the draw frame is:

A. 10%

B. 30%

C. 50%

D. Over 80% Ans: D

24. The critical drafting region for cotton is:

A. 1.15 to 1.4

B. 4 to 8

C. 10 to 12

D. 60 to 70

Ans: A

25. The pressure bar is located in the:

A. Creel zone

B. Main draft zone

C. Delivery zone

D. Coiler

Ans: B

26. The drafting wave is mainly caused by:

A. Excess twist

B. Hard top rollers

C. Uncontrolled movement of floating short fibres

D. High delivery speed only

Ans: C

27. A third draw frame passage can cause:

A. Quality deterioration

B. Large evenness gain

C. More dust removal

D. Higher draft range

Ans: A

Comber Preparatory and Comber

28. The number of machine passages needed between card and comber is:

A. 1

B. 2

C. 3

D. 5

Ans: B

29. The conventional comber lap sequence is:

A. Card → Draw frame → Lap former → Comber

B. Card → Comber → Draw frame

C. Card → Ribbon lap → Sliver lap → Comber

D. Card → Sliver lap → Ribbon lap → Comber

Ans: D

30. A super lap (lap former) can double up to:

A. 24 slivers

B. 32 slivers

C. 60 slivers

D. 100 slivers

Ans: C

31. The ribbon lap machine doubles:

A. 2 laps

B. 4 laps

C. 24 laps

D. 6 laps

Ans: D

32. Noil percentage in normal or ordinary combing is:

A. Up to 5%

B. 5–12%

C. 10–18%

D. Above 18%

Ans: C

33. Backward (counter) feed combers are used for:

A. Short staple fibres

B. Long staple fibres

C. Waste cotton

D. Rotor spinning

Ans: B

34. The periodic wave caused by comber piecing has a wavelength of:

A. 3–5 cm

B. 100 cm

C. 30–35 cm

D. 300 cm

Ans: C

35. Long fibres are pulled through the top comb by the:

A. Feed rollers

B. Calendar rollers

C. Lap rollers

D. Detaching rollers

Ans: D

36. Combed yarn is suitable for counts of about:

A. 4s–10s

B. 30s–120s Ne

C. 6s–24s

D. 1s–3s

Ans: B

Speed Frame

37. The twist range in roving is:

A. 5–10 TPM

B. 500–600 TPM

C. 30–65 TPM

D. 1000–2000 TPM

Ans: C

38. In the bobbin leading system:

A. Flyer speed is greater than bobbin speed

B. Bobbin speed is greater than flyer speed

C. Both speeds are equal

D. The bobbin is stationary

Ans: B

39. The draft range of the speed frame is:

A. 1–2

B. 5–22

C. 60–70

D. 100–200

Ans: B

40. The taper angle of the roving bobbin is:

A. 10°–20°

B. 45°–60°

C. 80°–95°

D. 120°–150°

Ans: C

41. Winding on the speed frame is produced by the:

A. Speed difference between flyer and bobbin

B. Traveler drag

C. Air blast

D. Rotor

Ans: A

42. Which spinning system does NOT need a speed frame?

A. Ring spinning

B. Rotor spinning

C. Compact spinning

D. Ring doubling

Ans: B

Ring Spinning

43. Twist in ring spinning is inserted by the:

A. Flyer

B. Rotor

C. Traveler

D. Air nozzle

Ans: C

44. The draft capability of a ring frame is up to about:

A. 10

B. 30

C. 70–75

D. 200

Ans: C

45. The spindle speed of a ring frame is:

A. 1,500–1,600 rpm

B. 5,000–10,000 rpm

C. 100,000 rpm

D. 18,000–22,000 rpm

Ans: D

46. The maximum traveler velocity is about:

A. 3–4 m/s

B. 15–20 m/s

C. 35–40 m/s

D. 100 m/s

Ans: C

47. The spinning triangle in ring spinning mainly increases:

A. Yarn hairiness

B. Yarn strength

C. Production speed

D. Twist efficiency

Ans: A

48. Ring spinning production speed is limited mainly by:

A. Roving count

B. Traveler friction and heat

C. Lappet guide

D. Creel type

Ans: B

Doubling

49. A TFO inserts how many twists per spindle revolution?

A. One

B. Two

C. Three

D. Four

Ans: B

50. Wet doubling is used for:

A. Tyre cord

B. Sailcloth

C. Sewing threads and voile yarns

D. Raw cotton

Ans: C

51. The equivalent twist of a TFO is about:

A. 500 TPM

B. 2,000 TPM

C. 8,000 TPM

D. 24,000–28,000 TPM

Ans: D

52. A TFO runs at 12,000 rpm with a delivery of 24 m/min. TPM is:

A. 500

B. 1,000

C. 2,000

D. 24,000

Ans: B

Non-Conventional Spinning

53. Rotor spinning production compared to ring spinning is:

A. Equal

B. 2–3 times

C. 6–10 times

D. 50 times

Ans: C

54. The commercial count range of rotor yarn is:

A. 6s–24s Ne

B. 30s–60s Ne

C. 80s–120s Ne

D. 150s–200s Ne

Ans: A

55. In air-jet spinning, the nozzle that inserts false twist into the core is:

A. N_1

B. N_2

C. Both equally

D. Neither

Ans: B

56. Strength of 100% cotton air-jet yarn compared to ring yarn is:

A. 50–60%

B. 100%

C. 120%

D. 200%

Ans: A

57. Air vortex spinning wraps fibre ends around a:

A. Rotating rotor

B. Traveler

C. Flyer

D. Stationary hollow spindle

Ans: D

58. Compact spinning can reduce yarn hairiness by up to:

A. 15%

B. 30%

C. 50%

D. 85%

Ans: D

Post-Spinning

59. Standard weight of a yarn bundle is:

A. 1 kg

B. 2.5 kg

C. 4.5 kg or 5 kg

D. 20 kg

Ans: C

60. The correct post-spinning sequence is:

A. Reeling → Conditioning → Baling → Bundling

B. Baling → Bundling → Reeling → Conditioning

C. Conditioning → Reeling → Baling → Bundling

D. Conditioning → Reeling → Bundling → Baling

Ans: D